

Ankush Makhijani

ankushgul.makhijani@sjsu.edu | +1 856-263-1347

<https://www.ankushmakhijani.com> | <https://www.linkedin.com/in/ankush-makhijani/> | <https://www.github.com/wolfy5786>

Education

San Jose State University | US

May 2027

Masters of Science | **Major: Software Engineering** | **Sp: AI & Distributed Systems**

Rutgers University, College of Arts and Science | US

May 2025

Bachelor of Science | **Major: Computer Science**

Granted with the Chancellor's Merit Scholarship at Rutgers University.

Dean's List

Work Experience

AceMyTravel (Stealth Startup) | Remote

June 2025 – October 2025

Software Engineer in AI | Python, FastAPI, Docker, SQL, GCP NLP, Lang chain

- Designed and prototyped an AI-powered itinerary generation system product using LangChain prompt flows and lightweight agents to convert user prompts and live travel data into personalized tours.
- Built gRPC-based microservices for booking updates and inter-service communication, reducing API latency by 70%..
- Integrated Google Cloud NLP APIs for semantic analysis and search, improving recommendation relevance by 28%.
- Working in cross-team collaboration teams to design, prototype, and deliver customer-facing features. Optimized performance with caching, async processing, and reactive design, reducing average response time by **19%**.

Simplified IT Solutions | Thane, India

April 2023 –August 2023

Software Intern | Java, MySQL, REST API, Spring Boot, Swagger

- My key role was in the development and maintenance of middle-tier services of ICICI pru bank using **Java with Spring Boot** and **REST APIs** to support a scalable web application architecture.
- Practiced Test-Driven Development (TDD) with JUnit and Mockito, improving release stability and reducing regressions.
- Validated data contracts between APIs and frontend components, detecting several critical bugs early in the development cycle, preventing production issues.

Skills

- AI & Machine Learning: Kira, LangGraph /LangSmith, TensorFlow, Pytorch, Keras, RAG, MCP, Semantic Search
- Programming Languages: Python, Java, C, JavaScript, SQL, TypeScript, GO
- Cloud: AWS (Lambda, DynamoDB, S3, EC2, RDS, API Gateway, CloudWatch), GCP (compute Engine, cloud SQL, NLP)
- Database: MySQL, PostgreSQL, MongoDB, Neo4j,
- Backend: Spring Boot, Node.js, Hibernate, Flask, Fast API, JDBC, Django, Redis, Kubernetes
- CI /CD: GitLab, Github Actions, Jenkins, Docker
- Software Testing: JUnit, Mockito, PyUnit, Selenium, Playwright

Research

Rutgers New Brunswick | New Brunswick, NJ

May 2024 – October 2024

Research Assistantship |C, Linear Programming, Data visualization

- Implemented polynomial approximations for log1p across all IEEE-754 rounding modes for 32-bit floats.
- Achieved 1.4× speedup and higher numerical accuracy compared to the standard libm implementation across 7M+ test inputs.
- Applied linear programming to iteratively retrain polynomials on failing constraints, improving robustness and precision.

Projects

Boost – Spec Driven AI assisted Software Testing | Python, LangGraph, LangSmith, AWS

December 25 – present

- Designed and built an AI-assisted software testing platform that generates production-ready test suites from natural language prompts and formal specifications.
- Implemented dense semantic search combined with AST-based code analysis to retrieve relevant source files, APIs, and user documentation across large, multi-module codebases.
- Designed a hybrid hierarchical + AST-backed indexing system that reduced retrieved file count by ~60% and context token size by 45% compared to embedding-only retrieval on large repositories (1k+ files)
- Orchestrated multi-step retrieval, generation, and validation pipelines using LangGraph, reducing end-to-end test generation latency by 30% through targeted context assembly.

Online Multiplayer Poker | Python, Django, Redis, MongoDB, Docker

September 25 – December 25

- Developed a real-time multiplayer poker game using **Django Channels** and **WebSockets** with **Redis** for scalable socket management, and built a responsive **React + Tailwind UI** with live updates and seamless player interaction.
- Integrated Multiplayer lobby, with in game chatting, game replays and logs.
- Containerized frontend, backend, and Redis services using **Docker** and orchestrated them with docker-compose.